A Modern Handbook of Physics

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Preface

The basic sciences and physics, in particular, are of prime importance today in the training of engineers for the various branches of the national economy. This has led to fundamental changes in recent years in the teaching of physics in engineer ing institutes, and to the students of other educational institutions in which they do not major in physics. The scope and scientific level of physics courses have been substantially supplemented and cover the main trends in the development of modern physics. Consequently, the physics textbooks for engineering students have inevitably become three-volume editions of almost fifteen hundred pages. The need has arisen, in this connection, for a concise handbook on this subject. The aim of the authors was to fulfill this need.

In scope and depth this handbook includes all of the definitions, formulas and information covered in the most comprehensive and up-to-date physics courses of engineering institutes and the physics departments of universities and colleges. Physical laws are concisely formulated, all the necessary explanations are given and, in many cases, derivations as well. Though it plays a vi-

tal role in the teaching of physics, experimental material has been omitted. This is due only to a lack of space. All the units and symbols comply with the requirements of the SI. Units of physical quantities and systems of units are listed and dealt with in a short appendix.

This handbook is designed primarily for engineering students, as well as college and university students studying, but not majoring in physics. It can be used to advantage by engineers and graduate students, as well as by instructors and lecturers of intermediate schools and colleges.

Mathematical knowledge required in using the handbook is within the scope of the ordinary mathematics courses of engineering institutes. The detailed index and the numerous cross references, indicating the chapter, section and subsection, are of aid in finding any required information. The authors are sincerely grateful to the translator, N. Weinstein, and the science editor, T. Baranovskaya, for their comprehensive and expert work in the publication of the English edition of this handbook, as well as for their valuable suggestions, which should facilitate its use by the reader.

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Parts One, Four, Five and the appendices were written by Prof. A. A. Detlaf; Parts Two, Three, Six, Seven and Eight, by Prof. B. M. Yavorsky.

Part I

Mechanics

1 Kinematics

1.1 Mechanical Motion. The Subject Matter of Mechanics

1.1.1

The simplest form of motion in nature is *mechanical motion*. It consists of changes in the relative positions of bodies or their parts in space in the course of time. The branch of physics dealing with the laws of mechanical motion is called *mechanics*. In a more narrow sense of the word, this is often under stood to mean *classical mechanics*, which deals with the motion of macroscopic bodies travelling at velocities a great many times less than that of light in free space. Classical mechanics is based on Newton's laws and is therefore frequently referred to as *Newtonian mechanics*. The laws of motion of bodies at velocities close to that of light in free space are taken up in *relativistic mechanics* (**??**), and the laws of motion of micro particles (for example, the electrons in atoms, molecules, crystals, etc.) are the subject matter of *quantum mechanics* (**??**).

1.1.2

Classical mechanics comprises three basic branches: statics, kinematics and dynamics. Statics deals with the laws for the composition of forces and the conditions for the equilibrium of bodies. *Kinematics* provides a mathematical description of all possible kinds of mechanical motion, regardless of what causes each particular kind of motion. *Dynamics* investigates the effect of the interaction between bodies on their mechanical motion.

1.1.3

The mechanical properties of bodies are determined by their chemical nature, internal structure and state. These are beyond the scope of mechanics and are studied in other branches of physics. Hence, in accordance with the conditions of each specific problem on hand, various simplified models are employed to describe real bodies in motion. These include the particle, the perfectly rigid body, the perfectly elastic body, the perfectly inelastic body and others.

A *material point*, or *particle*, is a body whose shape and size are of no consequence under the conditions of the problem being considered. For example, in a first approximation, the motion of a ship from one point to another can be dealt with as the motion of a particle. If, however, it becomes necessary to take into account such details of this motion as the pitching and rolling of the ship by swells or heavy seas, the ship should be dealt with as an extended body of definite shape. In the literature the word particle is more frequently used than material point.

Any extended body or system of such bodies, making up the *mechanical system* being investigated, can be regarded as a *system of particles*. For this purpose, ail the bodies of the system can be conceived of as being broken down into such a large number of parts that the size of each part is negligibly small compared to the sizes of the bodies themselves.

1.1.4

A *perfectly rigid body*, or simply *rigid body*, is one whose deformation can be neglected under the conditions of the given problem. The distance between any two points of a rigid body remains constant no matter what action the body is subjected to. A perfectly rigid body can be regarded as a system of particles rigidly connected to one another.

A *perfectly elastic body* is one whose deformation obeys Hooke's law (**??**). After the external forces applied to such a body have been removed, its initial size and shape are completely restored.

A *perfectly inelastic body* is one that completely retains its deformed state, caused by external forces applied to the body, when these forces have been removed.

1.2 Frames of Reference. Path, Path Length and Displacement Vector of a Particle

1.2.1

The position of a body in space can be specified only in reference to other bodies. It makes sense, for instance, to speak of the location of the planets with respect to the sun, that of an air liner or steamship with respect to the earth, but we cannot indicate their position in space "in general" without reference to some specific body. A perfectly rigid body in which a coordinate system is rigidly fixed, furnished with a "clock" and employed to determine the position of bodies and particles being investigated in space at various instants of time, is called a frame of reference. Sometimes the frame of reference is the coordinate system itself if it is provided with a "clock" and the solid in which the system is rigidly fixed is called the body of reference. In each specific problem the frame of reference is selected so as to simplify the problem to the maximum extent. Inertial frames of reference (2.1.2) are the most commonly used in physics.

1.2.2

Most extensively employed is the Cartesian rectangular system of coordinates (Figure 1.1) whose orthonormal (normal orthogonal) basis is formed by the three vectors $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$, which are unit vectors with respect to their magnitudes, mutually orthogonal and pass through the origin O of coordinates. The position of an arbitrary point M is specified by the radius vector $\mathbf{r}$, which connects the origin O of coordinates to point M. Vector $\mathbf{r}$ can be resolved along the axes $\mathbf{i}$, $\mathbf{j}$ and $\mathbf{k}$ of the basis as follows:

$$\mathbf{r} = x\mathbf{i} + y\mathbf{j} + z\mathbf{k}$$

where $x\mathbf{i}$, $y\mathbf{j}$ and $z\mathbf{k}$ are the components of vector $\mathbf{r}$ along the coordinate axes. The coefficients of resolution, x, y and z, are the Cartesian coordinates of point M and, due to the orthogonality of the vectors of the basis, the projections of radius vector $\mathbf{r}$ on the corresponding coordinate axes.

The motion of a particle is completely specified if three continuous and single-valued functions of the time t,

$$x = x(t),\ y = y(t) \text{ and } z = z(t),$$

are given to describe the changes in the coordinates of the particle with time.

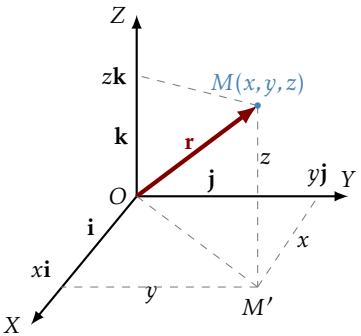


Figure 1.1: Cartesian system of coordinates.

These equations are said to be the *kinematic equations of motion of a particle*. They are equivalent to the single vector equation $\mathbf{r} = \mathbf{r}(t)$, for a moving particle.

1.2.3

The line described by a moving particle in space is called its *path*. The kinematic equations of motion of a particle specify the equation of its path in parametric form (the parameter being the time t). Depending upon the shape of the path, distinction is made between *rectilinear* and

curvilinear motion. The motion of a particle is said to be *plane* if its path lies completely in a single plane.

The mechanical motion of a body is relative, i.e. the nature of the motion and, in particular, the kind of path travelled by the points of the body depend upon the chosen frame of reference.

1.2.4

In the general case, the path of a moving particle is a spatial, rather than plane, curve. Introduced for such curves is the concept of an *osculating plane*. The osculating plane at the arbitrary point M of a curve is in the limiting position of a plane passing through any three points of the curve when these points approach point M without limit.

An *osculating circle* at point M of a curve is the limit of a circle passing through three points of the curve being considered when these points approach point M without limit. The osculating circle lies in the osculating plane. The centre of the osculating circle and its radius are called the *centre of curvature* and *radius of curvature*, at point M, of the curve being considered. The straight line connecting point M to the centre of curvature is called the *principal normal* to the curve at point M. The tangent to the curve at point M is perpendicular to the principal normal and also lies in the osculating plane.

1.2.5

The *path length* is the sum of the lengths of all the portions of the path over which the particle passes m the given interval of time. The instant of time $t = t_0$, from which we begin to consider the motion of a particle, is called the initial instant of time, and the position of the particle at this instant (point A in Figure 1.2) is called its initial position. Owing to the arbitrary nature of the time reference point it is usually assumed that $t_0 = 0$.

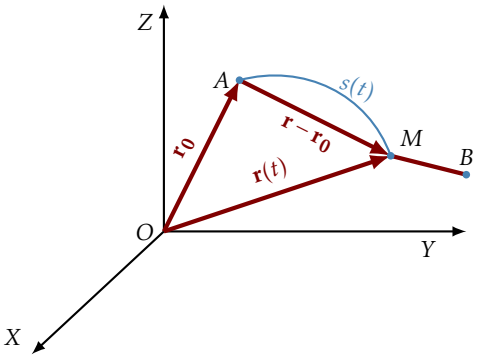


Figure 1.2: A particle moving on a curve.

The path length s, travelled by the particle from its initial position, is a scalar function of time: $s = s(t)$. As is obvious from its definition, the path length of a particle cannot be a negative quantity. If the particle travels along an arc of the path AB (Figure 1.2), always in the same direction and is at point M at the instant of time t,then $s(t) = \widehat{AM}$. If the particle travels along the path in a more complicated way, for example if it trav-

els from A to B, reaching B at the instant of time $t_1 < t$, and then reverses, returning to point M at instant t, then $s(t) = \widehat{AB} + \widehat{BM}$.

1.2.6

The displacement vector of a moving particle during the time interval from $t = t_1$ to $t = t_2$ is the vector from the, position of the particle at the instant of time t_1 to its position at instant t_2. It is equal to the increment in the radius vector of the particle during the time interval being considered:

$$\mathbf{r_2} - \mathbf{r_1} = \mathbf{r}(t_2) - \mathbf{r}t_1$$

The displacement vector is always directed along the chord subtending the corresponding portion of the path.

Shown in Figure 1.2 is the displacement vector of the particle for the time interval from t_0 to t. It is equal to $\mathbf{r} - \mathbf{r_0} = \mathbf{r}(t) - \mathbf{r}(t_0)$.

The displacement vector of a moving particle for the time interval from t to $t + \Delta t$ equals

$$\Delta r = r(t + \Delta t) - r(t) = \Delta x \times \mathbf{i} + \Delta y \times \mathbf{j} + \Delta z \times \mathbf{k}$$

where Δx, Δy and Δz are the increments (changes) in the coordinates of the particle during the time interval being considered.

1.2.7

A particle travelling freely in space can accomplish only three *independent motions*, i.e. motions that cannot be

represented as combinations of the other two. As a matter of fact, the motion of a particle along each of the axes of a Cartesian rectangular coordinate system cannot be obtained by its motion along the other two axes. The number of independent motions that a mechanical system can accomplish is called the *number of degrees of freedom* of the system. Hence, a free particle has three degrees of freedom.

1.3 Velocity

1.3.1

The concept of velocity is introduced in mechanics to characterize the rate of motion of bodies. It is the time rate of change in the position of moving bodies. The *average velocity* of a moving particle in the time interval from t to $t+\Delta t$ is the vector $\mathbf{v_{av}}$, equal to the ratio of the increment $\Delta\mathbf{r}$ in the radius vector of the particle during this time interval to the length of the time interval Δt:

$$\mathbf{v_{av}} = \frac{\Delta\mathbf{r}}{\Delta t}$$

Vector $\mathbf{v_{av}}$ has the same direction as $\Delta\mathbf{r}$, i.e. along the chord subtending the corresponding portion of the particle's path.

1.3.2

The *velocity* (or *instantaneous velocity*) of a moving particle is the vector quantity $\mathbf{v}$, equal to the first time deriva-

tive of the radius vector $\mathbf{r}$ of the particle being considered:

$$\mathbf{v} = \frac{\mathrm{d}\mathbf{r}}{\mathrm{d}t}$$

The velocity of the particle at the instant of time t is the limiting value that the average velocity $\mathbf{v_{av}}$ approaches for an infinitely small value of the time interval Δt. Thus

$$\mathbf{v} = \lim_{\Delta t \to 0} \frac{\Delta \mathbf{r}}{\Delta t} = \lim_{\Delta t \to 0} \mathbf{v_{av}}$$

The velocity vector $\mathbf{v}$ of the particle has the direction along a tangent to the path toward the motion in the same way as the vector $d\mathbf{r} = \mathbf{v}dt$ of infinitely small displacement of the particle during an infinitely short time interval dt.

The path length ds travelled by the particle during the length of time dt is equal to the magnitude of the displacement vector: $ds = |d\mathbf{r}|$. Therefore the magnitude of the velocity vector of a moving particle is equal to the first time derivative of the path length:

$$v = |\mathbf{v}| = \frac{\mathrm{d}s}{\mathrm{d}t}$$

1.3.3

The resolution of vector $\mathbf{v}$ along the basis of a Cartesian rectangular coordinate system is of the following form:

$$\mathbf{v} = v_x \times \mathbf{i} + v_y \times \mathbf{j} + v_z \times \mathbf{k}$$

The projections of the velocity of a moving particle on the coordinate axes are equal to the first time derivative

of the corresponding coordinates of the particle:

$$v_x = \frac{\mathrm{d}x}{\mathrm{d}t},\ v_y = \frac{\mathrm{d}y}{\mathrm{d}t},\ v_z = \frac{\mathrm{d}z}{\mathrm{d}t}$$

and the magnitude of the velocity vector, sometimes called the *speed*, is

$$v = |\mathbf{v}| = \sqrt{\left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}x}{\mathrm{d}t}\right)^2}$$

1.3.4

In rectilinear motion of a particle, the direction of its velocity vector remains constant. The motion of a particle is said to be *uniform* if the magnitude of its velocity does not change with time: $v = ds/dt = \text{const}$. The path length s travelled by a uniformly moving particle is a linear function of time: $s = vt$ (provided that $t_0 = 0$, see (1.2.5)).

If the magnitude of the particle's velocity, i.e. its speed, increases with time ($dv/dt > 0$), the motion is said to be *accelerated*; if the speed decreases with time ($dv/dt < 0$), the motion is said to be *decelerated*.

1.3.5

The *average velocity of nonuniform motion* of a particle over a given portion of its path is the scalar v_{av}, called the *travelling speed*, which is equal to the ratio of the length Δs of this portion to the length Δt of the time interval required to travel over this portion:

$$v_{\text{av}} = \frac{\Delta s}{\Delta t}$$

It is equal to the speed of uniform motion in which the time required to travel the same path length As is the same as in the nonuniform motion being considered.

In curvilinear motion of a particle $|\Delta\mathbf{r}| < \Delta s$. Therefore, in the general case, the average velocity (travelling speed) v_{av} of the particle is not equal to the magnitude of its average velocity $\mathbf{v_{av}}$ over the same portion of its path (1.3.1): $v_{\text{av}} \geqslant |\mathbf{v}|$ where the equality sign corresponds to a rectilinear portion of the path.

1.3.6

In the case of plane motion of a particle M (1.2.3), it often proves convenient to use the polar coordinates r and φ, where r is the distance from pole O to particle M and φ is the vectorial angle measured from polar axis OA (Figure 1.3). The velocity $\mathbf{v}$ of particle M can be resolved

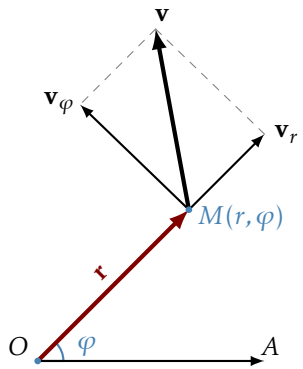


Figure 1.3: Polar coordinates of a moving particle.

into two mutually perpendicular components: the *radial velocity* $\mathbf{v}_r$ and the *transverse velocity* $\mathbf{v}_\varphi$:

$$\mathbf{v} = \mathbf{v}_r + \mathbf{v}_\varphi$$

in which

$$\mathbf{v}_r = \frac{1}{r}\frac{\mathrm{d}r}{\mathrm{d}t}\mathbf{r}, \text{ and } \mathbf{v}_\varphi = \frac{\mathrm{d}\varphi}{\mathrm{d}t}[\mathbf{kr}]$$

where $\mathbf{r}$ is the polar radius vector of particle M, and $\mathbf{k}$ is the unit vector directed perpendicular to the plane of motion of the particle so that the rotation of radius vector $\mathbf{r}$ upon an increase in angle φ is seen, from the head of unit vector $\mathbf{k}$, to be counterclockwise.

The magnitude v of the velocity vector of particle M, in plane motion, is

$$v = \sqrt{\left(\frac{\mathrm{d}r}{\mathrm{d}t}\right)^2 + \left(r\frac{\mathrm{d}\varphi}{\mathrm{d}t}\right)^2}$$

During the infinitely short length of time dt, polar radius vector $\mathbf{r}$ of the particle in plane motion sweeps over a sector of a circle, the area of the sector being $dS = r^2\,d\varphi/2$. Hence, the quantity

$$\sigma = \frac{\mathrm{d}S}{\mathrm{d}t} = \frac{1}{2}r^2\frac{\mathrm{d}\varphi}{\mathrm{d}t} = \frac{1}{2}rv_\varphi$$

is called the *areal* velocity.

1.4 Acceleration

1.4.1

The concept of acceleration is introduced in mechanics to characterize the rate of change of the velocity vector of a mov ing particle. The *average acceleration* in the time interval from t to $t + \Delta t$ is the vector $\mathbf{a}_{\text{av}}$, which is equal to the ratio of the increment $\Delta\mathbf{v}$ of the velocity vector of the particle, during the lime interval, to the length of the time interval Δt:

$$\mathbf{v}_{\text{av}} = \frac{\Delta\mathbf{v}}{\Delta t}$$

1.4.2

Acceleration (or *instantaneous acceleration*) of a moving particle is the vector quantity $\mathbf{a}$, equal to the first time derivative of the velocity $\mathbf{v}$ of the particle being considered or, other wise, the second time derivative of the particle's radius vector $\mathbf{r}$:

$$\mathbf{a} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2}$$

The acceleration of a moving particle at the instant of time t is the limiting value that the average acceleration $\mathbf{a}_{\text{av}}$ approaches for an infinitely small time interval Δt. Thus

$$\mathbf{a} = \lim_{\Delta t \to 0} \frac{\Delta\mathbf{v}}{\Delta t} = \lim_{\Delta t \to 0} \mathbf{a}_{\text{av}}$$

1.4.3

The resolution of vector $\mathbf{a}$ along the basis of a rectangular Cartesian coordinate system is of the form:

$$\mathbf{a} = a_x\mathbf{i} + a_y\mathbf{j} + a_z\mathbf{k}$$

The projections of the acceleration vector of a moving particle on the coordinate axes are equal to the first time derivatives of the corresponding velocity projections or, otherwise, the second time derivatives of the corresponding coordinates of the particle:

$$a_x = \frac{\mathrm{d}v_x}{\mathrm{d}t} = \frac{\mathrm{d}^2x}{\mathrm{d}t^2},\ a_y = \frac{\mathrm{d}v_y}{\mathrm{d}t} = \frac{\mathrm{d}^2y}{\mathrm{d}t^2},\ a_z = \frac{\mathrm{d}v_z}{\mathrm{d}t} = \frac{\mathrm{d}^2z}{\mathrm{d}t^2}$$

The magnitude of the acceleration vector is

$$a = |a| = \sqrt{\left(\frac{\mathrm{d}v_x}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}v_y}{\mathrm{d}t}\right)^2 + \left(\frac{\mathrm{d}v_z}{\mathrm{d}t}\right)^2}$$
$$= \sqrt{\left(\frac{\mathrm{d}^2x}{\mathrm{d}t^2}\right)^2 + \left(\frac{\mathrm{d}^2y}{\mathrm{d}t^2}\right)^2 + \left(\frac{\mathrm{d}^2z}{\mathrm{d}t^2}\right)^2}$$

1.4.4

The acceleration vector of a moving particle lies in the osculating plane (1.2.4) passing through point M of the path being considered and is direct ed toward the concavity of the path BC (Figure 1.4). In this plane the acceleration vector $\mathbf{a}$ can be resolved into two mutually perpendicular components $\mathbf{a}_\tau$ and $\mathbf{a}_n$:

$$\mathbf{a} = \mathbf{a}_\tau + \mathbf{a}_n$$

1.4.5

Component $\mathbf{a}_\tau$ is called the *tangential acceleration* of the moving particle. It is directed along the tangent to the

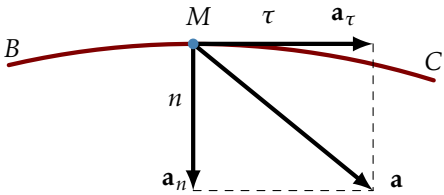


Figure 1.4: Tangential and normal components of the acceleration of a moving particle.

path and equals

$$\mathbf{a}_\tau = \frac{\mathrm{d}v}{\mathrm{d}t}\tau \text{ and } a_t = \frac{\mathrm{d}v}{\mathrm{d}t}$$

where $\tau = \mathbf{v}/v$ is the unit vector of the tangent, from point M of the path in the direction of velocity $\mathbf{v}$ of the particle, and a_τ is the projection of the tangential acceleration on the direction of vector $\mathbf{v}$. The tangential acceleration characterizes the rate of change of the magnitude of the particle's velocity vector. Vectors $\mathbf{a}_\tau$ and $\mathbf{v}$ coincide in direction, i.e. $a_\tau > 0$, in accelerated motion (1.3.4); vectors $\mathbf{a}_\tau$ and $\mathbf{v}$ are opposite in direction $a_\tau < 0$, in decelerated motion of the particle; and $a_\tau = 0$ in its uniform motion. If $a_\tau = \text{const} \neq 0$, the motion is said to be uniformly variable. In uniformly variable motion the magnitude of the particle's velocity is linearly time-dependent:

$$v = v_0 + a_\tau t$$

where $v_0 = v(0)$ is the magnitude of the initial velocity, i.e. the velocity at the initial instant of time $t = 0$. If $a_\tau = \text{const} > 0$, the motion of the particle is said to be

uniformly accelerated, whereas if $a_\tau = \text{const} < 0$, it is said to be *uniformly decelerated*.

1.4.6

Component $\mathbf{a}_n$ of acceleration $\mathbf{a}$ of the moving particle is called its *normal acceleration*. It is directed along the principal normal to the path at point M being considered, toward the centre of curvature (1.2.4) of the path. Therefore, an is often called the *centripetal acceleration* of the particle. The normal acceleration equals

$$\mathbf{a}_n = \frac{v^2}{R}\mathbf{n}$$

where $\mathbf{n}$ is the unit vector of the principal normal, and R is the radius of curvature of the path. The normal acceleration characterizes the rate of change in direction of the particle's velocity vector.

If the particle travels in a straight line the normal acceleration $\mathbf{a}_n = 0$ and the acceleration is equal to the tangential acceleration of the particle: $\mathbf{a} = \mathbf{a}_\tau$.

1.5 Translational and Rotary Motion of a Rigid Body[1]

1.5.1

Translation is the kind of motion of a rigid body in which any straight line rigidly fixed in the body (for instance,

[1] Only perfectly rigid bodies are dealt with in this section; for the sake of brevity, they have been called simply "rigid bodies".

line AB in Figure 1.5), remains parallel to its initial position (A_0B_0). With respect to the earth, a lift cabin (elevator car), lathe tool, and many other items have translational motion. In the translational motion of a rigid rigid

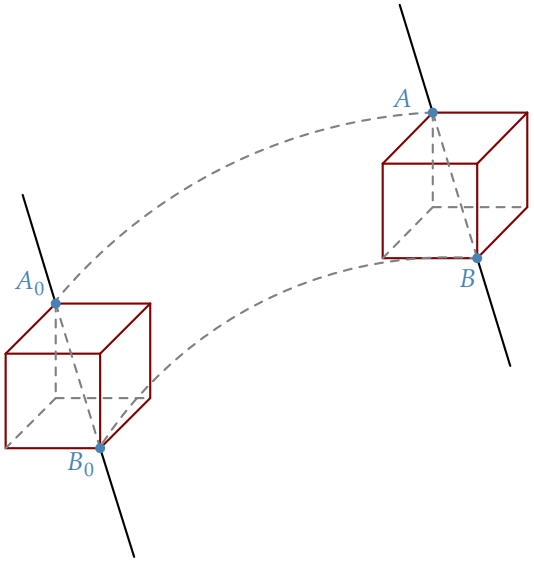


Figure 1.5: Translational motion of a rigid body.

body all of its points are displaced in an identical way.

During the infinitely short time dt the radius vectors of these points are changed by the same amount dr. Hence the velocity of all points of the body is the same in each instant of time and is equal to d$\mathbf{r}$/dt and consequently, their accelerating is also the same. For this reason, a kinematic discussion of the translational mo-

tion of a rigid body reduces to the investigation of the motion of any one of its points, or particles. Dynamics usually deals with the motion of the centre of mass of the body (2.3.3). A rigid body travel ling freely in space has three translational degrees of freedom (1.2.7), corresponding to its translational motions along the three coordinate axes.

1.5.2

The motion of a rigid body in which two of its points, A and B, remain fixed is called *rotation (or rotary motion) of the body about a fixed axis*. The fixed straight line AB is the axis of rotation. In the rotation of a rigid body about a fixed axis all the points of the body describe circles whose centres lie on the axis of rotation and whose planes are perpendicular to this axis. The rotors of turbines, motors and generators, for example, have rotary motion with respect to the earth if these machines are rigidly mounted.

A rigid body rotating about a fixed axis has only one degree of freedom (1.2.7). Its position in space can be completely defined by specifying the angle of rotation (position angle) φ from some definite (initial) position.

1.5.3

The angular velocity serves to characterize the rate and direction of rotation of a body about an axis. The *angular velocity* is the vector $\boldsymbol{\omega}$, which is equal in magnitude to the first derivative of the angle of rotation φ with respect to time t and is directed along the fixed axis of rotation

in such a way that from its head the rotation of the body is counterclockwise (Fig. 1.6)[2]:

$$\boldsymbol{\omega} = \frac{\mathrm{d}\boldsymbol{\varphi}}{\mathrm{d}t} \text{ and } \omega = \frac{\mathrm{d}\varphi}{\mathrm{d}t}$$

Here $\mathrm{d}\boldsymbol{\varphi}$ is the vector of an element of rotation of the body during the time dt. This vector is directed along the axis of rotation, also according to the right-hand screw rule (Figure 1.6). Axial vectors $\mathrm{d}\boldsymbol{\varphi}$ and $\boldsymbol{\omega}$ have no definite points of application: they can be laid off from any point on the axis of rotation. In Figure 1.6 they are laid off from a certain point O of the fixed axis of rotation. This point is also taken as the origin of the reference frame.

Rotation of a body is said to be *uniform* if the magnitude of its angular velocity remains constant with time $\omega = \text{const}$. In this case the angle of rotation of the body is linearly dependent on time: $\varphi = \omega t$.

1.5.4

An arbitrary point M of a rigid body rotating about fixed axis OZ with the angular velocity $\boldsymbol{\omega}$ describes a circle of radius ρ with its centre at point O' (Figure 1.7). As distinct from the angular velocity of the body, the velocity

[2] The direction of vector $\boldsymbol{\omega}$ can also be determined by the *right-hand screw rule*. It coincides with the direction of transla tional motion of a right-hand screw rotating together with the body. Vectors like $\boldsymbol{\omega}$, whose direction is related to the direction of rotation and is reversed in transforming from a right-handed to a left-handed system of coordinates, are called *pseudovectors*, or *axial vectors* (in distinction to ordinary, *polar vectors*, which are not reversed in the above-mentioned transformation of coor dinates). For example, the vector product of two polar vectors is a pseudovector, and that of a pseudovector and a polar vector is a polar vector.

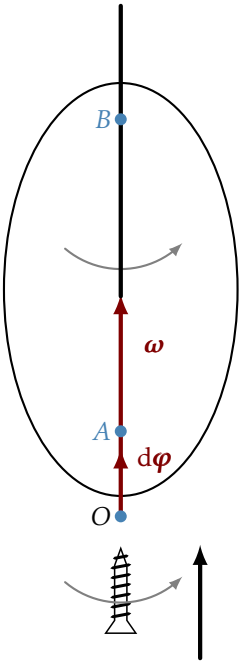


Figure 1.6: Angular velocity and right-hand rule.

v of point M is often called the *linear velocity*. It is perpendicular both to the axis of rotation (i.e. to vector $\boldsymbol{\omega}$) and to radius vector $\boldsymbol{\rho}$, from the centre O' of the circle to point M, and is equal to their vector product:

$$\mathbf{v} = [\boldsymbol{\omega}\boldsymbol{\rho}] = [\boldsymbol{\omega}\mathbf{r}] \text{ and } v = \omega\rho$$

Here $\mathbf{r} = \mathbf{OO'} + \boldsymbol{\rho}$ and is the radius vector of point M, from point O of the axis of rotation. Point O is taken as

the origin of coordinates.

1.5.5

The *period of revolution* is the time interval T required for a body, rotating uniformly at the angular velocity ω, to complete one revolution about its axis of rotation (i.e. to rotate through the angle $\varphi = 2\pi$):

$$T = \frac{2\pi}{\omega}$$

The *rotational speed* $n = 1/T = \omega/2\pi$ is the number of revolutions made by a body in unit time in uniform rotation at the angular velocity ω.

1.5.6

The rotation of a rigid body in which one of its points remains fixed is said to be *rotation about a fixed point*. This point is usually taken as the origin of the fixed reference frame. In such rotation, all the points of the body travel along the surfaces of concentric spheres whose centre is at the fixed point.

At each instant of time, this motion of the rigid body can be regarded as rotation about a certain axis that passes through the fixed point and is called the instantaneous axis of rotation. In the general case, the position of the *instantaneous axis of rotation* varies with respect to both a fixed reference frame and one rigidly fixed in the rotating body.

The velocity $\mathbf{v}$ of arbitrary point M of such a body is

$$\mathbf{v} = [\boldsymbol{\omega}\mathbf{r}] \text{ and } v = \omega\rho.$$

Here $\boldsymbol{\omega} = \mathrm{d}\boldsymbol{\varphi}/\mathrm{d}t$ is the angular velocity of the body and is directed along the instantaneous axis of rotation as is the vector $\mathrm{d}\boldsymbol{\varphi}$ of an element of rotation of the body during the infinitely short time $\mathrm{d}t$, $\mathbf{r}$ is the radius vector from fixed point O, about which the body rotates, to point M, and ρ is the distance from point M to the instantaneous axis of rotation. Such a body can have three independent motions: it can rotate about each of three mutually perpendicular axes passing through fixed point O. It consequently has three degrees of freedom (1.2.7).

1.5.7

The *angular acceleration* vector $\boldsymbol{\epsilon}$ is introduced to indicate the rate of change of the angular velocity vector with time in nonuniform rotation of a body about a fixed axis or its rotation about a fixed point. Vector $\boldsymbol{\epsilon}$ is equal to the first derivative of the angular velocity $\boldsymbol{\omega}$ of the body with respect to the time t:

$$\boldsymbol{\epsilon} = \frac{\mathrm{d}\boldsymbol{\omega}}{\mathrm{d}t}$$

If the body rotates about a fixed axis, vector $\boldsymbol{\epsilon}$ is directed along this axis. It coincides in direction with $\boldsymbol{\omega}$ in accelerated rotation ($\mathrm{d}\omega/\mathrm{d}t > 0$) and is opposite to $\boldsymbol{\omega}$ in direction in decelerated rotation ($\mathrm{d}\omega/\mathrm{d}t < 0$). The projection of the angular acceleration on the fixed axis OZ of rotation is equal to

$$\epsilon_z = \frac{\mathrm{d}\omega_z}{\mathrm{d}t}$$

where ω_z is the projection of vector $\boldsymbol{\omega}$ on the same axis.

1.5.8

As distinct from the angular acceleration, the acceleration a of arbitrary point M of a rigid body rotating about fixed point O, or about a fixed axis passing through point O, is often called the *linear acceleration*. It equals

$$\mathbf{a} = \frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \frac{\mathrm{d}}{\mathrm{d}t}[\boldsymbol{\omega}\mathbf{r}] = \mathbf{a}_r + \mathbf{a}_{\mathrm{ax}}$$

where $\mathbf{a}_r = [\boldsymbol{\epsilon}\mathbf{r}]$ is the *rotary acceleration* of the point, and $\mathbf{a}_{\mathrm{ax}} = [\boldsymbol{\omega}[\boldsymbol{\omega}\mathbf{r}]]$ is its *axipetal acceleration*, which is directed toward the instantaneous axis of rotation. If the body ro-

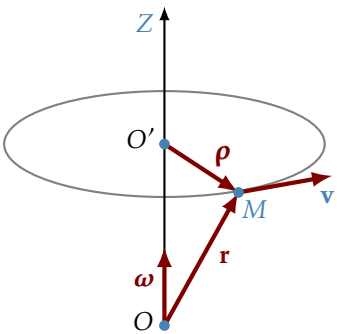


Figure 1.7: Relation between linear and rotational acceleration.

tates about the fixed axis OZ (Figure 1.7), the rotary acceleration of point M coincides with its tangential acceleration $\mathbf{a}_\tau$ (1.4.5), and its axipetal acceleration coincides with its normal acceleration (1.4.6):

$$\mathbf{a}_\tau = [\boldsymbol{\epsilon}\mathbf{r}] = [\boldsymbol{\epsilon}\boldsymbol{\rho}] \text{ and } \mathbf{a}_n = -\boldsymbol{\omega}^2\boldsymbol{\rho}$$

1.5.9

Any complex motion of a rigid body can be broken down into two simple motions: translational motion at the velocity $\mathbf{v}_A$ of a certain arbitrarily chosen point A of the body and rotation about an instantaneous axis passing through this point. The angular velocity $\boldsymbol{\omega}$ of rotation is independent of the choice of point A. The velocity of arbitrary point M of the body is

$$\mathbf{v} = \mathbf{v}_A + [\boldsymbol{\omega}(\mathbf{r} - \mathbf{r}_A)]$$

where $\mathbf{r}$ and $\mathbf{r}_A$ are the radius vectors of points M and A.

In the dynamics of a rigid body it usually proves convenient to deal with its complex motion as the combination of two simultaneous motions: translational motion at the velocity of the body's centre of mass (2.3.3) and rotation about the centre of mass.

The simplest case of complex motion of a body is *plane*, or *plane-parallel*, *motion*, in which all the points of the body travel in parallel planes. This is the motion, for instance, of a homogeneous circular cylinder rolling down an inclined plane. In plane motion the direction of the instantaneous axis of rotation of the body about point A remains constant and vectors $\boldsymbol{\omega}$ and $\mathbf{v}_A$ are perpendicular to each other.

2 Newton's Laws

2.1 Newton's First Law. Inertial Frames of Reference

2.1.1

As the *first law of dynamics* Newton took a law that had already been established by Galileo: a particle persisting its state of rest or uniform motion in a straight line until the action of other bodies compel it to change that state.

Newton's first law indicates that no external action is required to maintain a state of rest or of uniform motion in a straight line. This manifests a special dynamic property of bodies, called their *inertia*. Newton's first law is consequently called the *law of inertia*, and the motion of a body not subject to the action of other bodies is said to be *inertial motion*.

2.1.2

Mechanical motion is relative: its nature may differ for different frames of reference (1.2.1) that are moving in relation to each other. An astronaut, for instance, on board a man-made satellite orbiting the earth, may be at rest in a reference frame fixed in the satellite. At the same time, he is travelling together with the satellite with respect to the earth along an elliptic orbit, i.e. not with uniform motion and not in a straight line. It is thus obvious that Newton's first law does not hold for any frame of reference. A ball, for example, lying on the smooth floor of a stateroom in a ship travelling with uniform motion at sea in a straight line, may suddenly start moving along the floor without being subject to the action of any other bodies. This will occur when the velocity of the ship, i.e. its speed and/or direction, begins to change.

A frame of reference with respect to which a particle, not subject to external action, remains at rest or moves uniformly in a straight line is called an *inertial frame of reference*. The content of Newton's first law reduces, in essence, to two statements: in the first place, that all material bodies possess the property of inertia, and secondly, that inertial reference frames exist.

2.1.3

Any two inertial reference frames can move only with translational motion with respect to each other, and only uniformly in a straight line. It has been experimentally established that the *heliocentric frame of reference* is practically inertial. The origin of its coordinate system is located at the centre of mass (2.3.3) of the solar system (ap-

proximately at the centre of the sun) and the three axes extend in the directions of three distant stars selected, for example, so that these coordinate axes are mutually perpendicular.

A *laboratory frame of reference*, whose coordinate axes are rigidly fixed in the earth, is noninertial chiefly owing to the earth's diurnal (daily) rotation. But the earth rotates so slowly that the maximum normal acceleration (1.4.6) of points on its surface due to diurnal rotation does not exceed $0.034\,\mathrm{m/s^2}$. Therefore, in the great majority of practical problems, the laboratory reference frame can be regarded as being approximately inertial.

2.1.4

Inertial reference frames are of prime importance in all branches of physics, not only mechanics. This is due to the fact that according to Einstein's principle of relativity (**??**) the mathematical expression of any physical law should have the same form in ail inertial reference frames. Hence, in the following we shall make use of only inertial frames of reference, without stipulation each time. The laws of motion of a particle with respect to a noninertial reference frame is dealt with in Chapter **??**.

2.2 Force

2.2.1

Force is a vector quantity which is a measure of the mechanical action exerted on the body being considered by other

bodies. Mechanical interaction can be accomplished either be tween bodies in direct contact (for instance, in friction, and in the pressure of bodies against one another), or between remote bodies. A special form of matter, connecting particles of matter into unified systems and transmitting the action of certain particles on others at finite velocities, is called a *physical field*, or simply *field*. Interaction between remote bodies is accomplished by the gravitational or electromagnetic fields they set up (for example, the attraction of the planets to the sun, and the inter action of charged bodies or current-carrying conductors). Mechanical action exerted on a given body by other bodies can be manifested in two ways. In the first place, it can lead to a change in the state of mechanical motion of the body being considered, and secondly, to its deformation. Both of these elects can serve as the basis for measuring forces. For example, the measurement of forces by means of a spring dynamometer is based on Hooke's law (**??**) for axial tension.

Using the concept of force, we usually speak in mechanics of the motion and deformation of a body by the action of the forces applied to it. Here, of course, each force always corresponds to a certain body exerting this force on the body being considered.

Force $\mathbf{F}$ is completely specified by giving its magnitude, direction in space and point of application. The straight line along which a force is exerted is called its *line of action*.

A field acting on a particle with the force $\mathbf{F}$ is said to be *steady*, or *stationary*, if it does not change in the course of time t, i.e. if force $\mathbf{F}$ is explicitly independent of time at any point in the held: $\partial\mathbf{F}/\partial t \equiv 0$. For a held

to be steady it is necessary that the bodies which set it up are at rest with respect to the reference frame used in considering the held.

2.2.2

The simultaneous action of several forces, $\mathbf{F_1}, \mathbf{F_2}, \mathbf{F_3}, \ldots \mathbf{F_n}$ (Figure 2.1 (a)) on a particle M is equivalent to the action of a single force, called the *resultant force* and equal to the vector sum of the component forces. Thus

$$\mathbf{F} = \sum_{i=1}^{n} \mathbf{F_i}$$

It is the closing line of the polygon of forces $\mathbf{F_1}, \mathbf{F_2}, \mathbf{F_3}, \ldots \mathbf{F_n}$ (Figure 2.1 (b)).

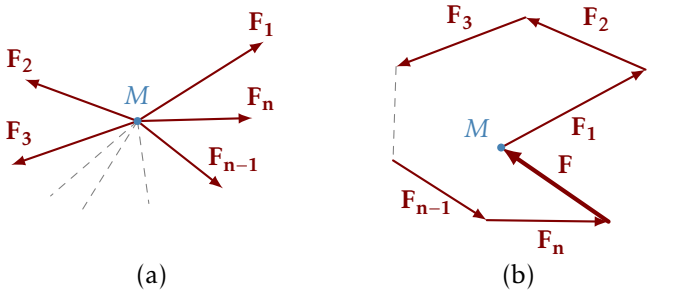


Figure 2.1: Resultant force is equal to vector sum of the component forces.

If a body is perfectly rigid, the action of a force on it does not change when the point of application is displaced along the line of action, but within the limits of

the body. In other words, forces applied to perfectly rigid bodies can be regarded as non-localized vectors.

2.2.3

A body is said to be *free* if no limitations are imposed on its position and motion in space. An airplane, for example, Hying in the air is a free body, as is a submarine travelling in the depths of the sea. Most bodies that we deal with are not free; certain limitations are imposed on their possible positions and motion. These limitations are called *constraints* in mechanics. A ball, for example, hung on a nonextensible thread, can not move away from its point of suspension by more than the length of the thread; a streetcar (tram) must travel only along its tracks. Constraints are realized due to the action on the body being considered exerted by other bodies attached to or contacting this body (for instance, the thread on the ball tied to it, the track on the streetcar, etc.).

In investigating the behaviour of constrained bodies or systems of bodies in mechanics, the *principle of releasability* is applied. It states that a restrained body (or system of bodies) can be treated as a free body (or system) if the bodies imposing the constraints are replaced by the corresponding forces. These forces are called *reaction forces* of the constraints, or *constraint reactions*. All other forces exerted on the body are called *active forces*. Thus, the motion of the ball hung by the thread can be treated as the motion of a free ball subject, in addition to all the active forces applied to it (for instance, gravity), to the reaction of the thread as well.

In contrast to the active forces, which should he given

in each specific problem, the constraint reactions are not known beforehand. They are to be determined in solving the problem. Their values should be such that under the combined effects of the active forces and constraint reactions, the motion of the "released" body completely agrees with the limitations imposed by the constraints on the constrained body being considered. There are no other distinctions between constraint reactions and active forces.

2.2.4

Bodies not included in the mechanical system being in vestigated are called *external bodies*. Forces exerted on the system by external bodies are called *external forces*. Correspondingly, forces of interaction between the parts or particles of the system being considered are called *internal forces*.

A mechanical system is said to be *closed*, or *isolated*, if it does not interact with external bodies. No external forces act on any of the bodies of a closed system.

2.3 Mass. Momentum

2.3.1

In classical (Newtonian) mechanics, the *mass of a particle* (material point) is a positive scalar quantity which is a measure of the inertia of this particle. Under the action of a force the velocity of a particle changes *gradually*, not instantaneously, i.e. it acquires an acceleration *finite* in magnitude and inversely proportional to the mass of the

particle. To compare the masses, m_1 and m_2, of two particles, it is sufficient to measure the magnitudes a_1 and a_2 of the accelerations acquired by these particles when the same force is applied to them: $m_2/m_1 = a_1/a_2$. The mass of a body is ordinarily determined by weighing by beam scales.

Classical (Newtonian) mechanics contends that:

(a) the mass of a particle is independent of its state of motion, being its unalterable characteristic;

(b) mass is an additive quantity, i.e. the mass of a system (for instance, a body) is equal to the sum of the masses of all the particles making up the system;

(c) the mass of an isolated (closed) system (2.2.4) does not change with any processes occurring within the system (*law of conservation of mass*).

These propositions of Newtonian mechanics were reconsidered and revised in relativistic mechanics (**??**, **??**, **??** and **??**).

2.3.2

The density ρ of a body at a given point M is the ratio of the mass dm of an infinitely small element of the body, which includes point M, to the quantity dV, which is the volume of this element. Thus

$$\rho = \frac{dm}{dV}$$

The size of the element being considered should be so small that any variation of density within its limits can

be neglected. On the other hand, this size should he a great many times larger than intermolecular distances.

A body is said to be *homogeneous* if the density is constant at all of its points. The mass of a homogeneous body is the product of its density by its volume: $m = \rho V$.

The mass of a nonhomogeneous body is

$$m = \int_{(V)} \rho dV$$

where ρ is a function of the coordinates, and integrating is carried out over the whole volume of the body. The average density $\langle \rho \rangle$ of a nonhomogeneous body is the ratio of its mass to its volume: $\langle \rho \rangle = m/V$.

2.3.3

The centre of inertia, or centre of mass, of a system of particles is point C, whose radius vector $\mathbf{r}_c$ equals

$$\mathbf{r}_C = \frac{1}{m} \sum_{i=1}^{n} m_i \mathbf{r}_i$$

where m_i and $\mathbf{r}_i$ are the mass and radius vector of the ith particle, n is the total number of particles in the system, and $m = \sum_{i=1}^{n} m_i$ is the mass of the whole system.

The velocity of the centre of mass is

$$\mathbf{v}_C = \frac{\mathrm{d}\mathbf{r}_C}{\mathrm{d}t} = \frac{1}{m} \sum_{i=1}^{n} m_i \mathbf{v}_i$$

2.3.4

The vector quantity $\mathbf{p}_i$, equal to the product of the mass m_i of a particle by its velocity $\mathbf{v}_i$, is called the *momentum of this particle*. The momentum of a system of particles is the vector $\mathbf{p}$, which is the geometric sum of the momenta of all the particles making up the system:

$$\mathbf{p} = \sum_{i=1}^{n} \mathbf{p}_i$$

The momentum of a system is equal to the product of the mass of the whole system by the velocity of its centre of mass: $\mathbf{p} = m\mathbf{v}_C$.

2.4 Newton's Second Law

2.4.1

The basic law of the dynamics of a particle is Newton's second law, which discusses how the mechanical motion of a particle is changed by the effect of forces applied to it. *Newton's second law* states: the rate of change in the momentum p of a particle is equal to the force $\mathbf{F}$ acting on it. Thus

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t} = \mathbf{F} \text{ or } \frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{v}) = \mathbf{F}$$

where m and $\mathbf{v}$ are the mass and velocity of the particle.

If several forces act simultaneously on a particle, the force $\mathbf{F}$ in Newton's second law is understood to he the geometric sum of all acting forces, both active forces and constraint reactions (2.2.3), i.e. the resultant force (2.2.2).

2.4.2

The vector quantity $\mathbf{F}\mathrm{d}t$ is called an element of *impulse of force* $\mathbf{F}$ during the infinitely short time $\mathrm{d}t$ of its action. The impulse of force $\mathbf{F}$ acting in the finite time interval from $t = t_1$ to $t = t_2$ is equal to the definite integral $\int_{t_1}^{t_2} \mathbf{F}\mathrm{d}t$, where $\mathbf{F}$, in the general case, depends upon the time t.

According to Newton's second law, the change in the momentum of a particle is equal to the impulse of the force acting on it:

$$\mathrm{d}\mathbf{p} = \mathbf{F}\mathrm{d}t = \Delta\mathbf{p} = \mathbf{p}_2 - \mathbf{p}_1 = \int_{t_1}^{t_2} \mathbf{F}\mathrm{d}t$$

where $\mathbf{p}_2 = \mathbf{p}(t_2)$ and $\mathbf{p}_1 = \mathbf{p}(t_1)$ are the values of the momentum of a particle at the end $t = t_2$ and beginning $t = t_1$ of Ihe time interval being considered.

2.4.3

Since in Newtonian mechanics the mass m of a particle is independent of its state of motion, $\mathrm{d}m/\mathrm{d}t = 0$. Hence, the mathematical expression of Newton's second law can also be presented in the form

$$\mathbf{a} = \frac{\mathbf{F}}{m}$$

where $\mathbf{a} = \mathrm{d}\mathbf{v}/\mathrm{d}t = \mathrm{d}^2\mathbf{r}/\mathrm{d}t^2$ is the acceleration of the particle and $\mathbf{r}$ is its radius vector. The corresponding statement of *Newton's second law reads*: the acceleration of a particle coincides in direction with the force acting on

it and is equal to the ratio of this force to the mass of the particle. The tangential and normal accelerations of a particle (1.4.4, 1.4.5 and 1.4.6) are determined by the corresponding components of force **F**:

$$\mathbf{a}_\tau = \frac{\mathbf{F}_\tau}{m},\ a_\tau = \frac{\mathrm{d}v}{\mathrm{d}t} = \frac{F_\tau}{m}$$

and

$$\mathbf{a}_n = \frac{\mathbf{F}_n}{m},\ a_n = \frac{v^2}{R} = \frac{F_n}{m}$$

where v is Ihe magnitude of the particle's velocity vector, and R is the radius of curvature of its path. The force F_n, imparting normal acceleration to a particle, is directed toward the centre of curvature (1.2.4) of the particle's path and is therefore called the *centripetal force*.

2.4.4

If a particle is subject to the simultaneous action of sev eral forces, $\mathbf{F_1}, \mathbf{F_2}, \mathbf{F_3}, \ldots \mathbf{F_n}$, its acceleration is

$$\mathbf{a} = \frac{1}{m}\sum_{l=1}^{n}\mathbf{F}_l = \sum_{l=1}^{n}\mathbf{a}_l$$

where $\mathbf{a}_l = \mathbf{F}_l/m$. Consequently, each of the forces acting simultaneously on a particle imparts to it the acceleration it would have if there were no other forces (*principle of superposition, or independence, of forces*).

The equation

$$m\frac{\mathrm{d}^2\mathbf{r}}{\mathrm{d}t^2} = \mathbf{F} = \sum_{l=1}^{n}\mathbf{F}_l$$

is called the *differential equation of motion of a particle*.

In projections on the axes of a rectangular Cartesian coordinate system, this equation is of the form

$$m\frac{\mathrm{d}^2x}{\mathrm{d}t^2} = F_x,\ m\frac{\mathrm{d}^2y}{\mathrm{d}t^2} = F_y,\ m\frac{\mathrm{d}^2z}{\mathrm{d}t^2} = F_z$$

where x, y and z are the coordinates of the moving particle.

2.5 Newton's Third Law. Motion of the Centre of Mass

2.5.1

The mechanical action exerted by bodies on one another is of the nature of their *interaction*. This is what *Newton's third law* deals with: two particles exert forces on each other which are equal in magnitude and opposite in direction along a line connecting these particles.

If $\mathbf{F}_{ik}$ is a force exerted on the ith particle by the kth one, and $\mathbf{F}_{ki}$ is, the force exerted on the kth particle by the ith one, then, according to Newton's third law,

$$\mathbf{F}_{ki} = -\mathbf{F}_{ik}$$

Forces $\mathbf{F}_{ik}$ and $\mathbf{F}_{ki}$ are applied to different particles and can be mutually counterbalanced only when particles i and k belong to the same perfectly rigid body.

2.5.2

Newton's third law is an essential supplement to the first and second laws. It permits us to go over from the dynamics of a separate particle to that of an arbitrary mechanical system (system of particles). It follows from Newton's third law that in any mechanical system the geometric sum of all the internal forces (2.2.4) equals zero:

$$\sum_{i=1}^{n}\sum_{k=1}^{n}\mathbf{F}_{ik}=0$$

where n is the number of particles in the system, and $\mathbf{F}_{ii}=0$. The vector $\mathbf{F}^{\text{ext}}$, equal to the vector sum of all the external forces (2.2.4) acting on the system, is called the *principal vector of external forces*:

$$\mathbf{F}^{\text{ext}}=\sum_{i=1}^{n}\mathbf{F}_{i}^{\text{ext}}$$

where $\mathbf{F}_{i}^{\text{ext}}$ is the resultant external force applied to the ith particle.

2.5.3

It follows from Newton's first and second laws that the first derivative with respect to the time t of the momentum $\mathbf{p}$ of a mechanical system (2.3.4) is equal to the principal vector of all external forces applied to the system. Thus

$$\frac{\mathrm{d}\mathbf{p}}{\mathrm{d}t}=\mathbf{F}^{\text{ext}}$$

This equation expresses the *law of the change in the momentum of a system*.

Since $\mathbf{p} = m\mathbf{v}_c$, where m is the mass of a system and $\mathbf{v}_c$ is the velocity of its centre of inertia, or mass, the *law of motion of the centre of mass* of a mechanical system is of the form

$$\frac{\mathrm{d}}{\mathrm{d}t}(m\mathbf{v}_c) = \mathbf{F}^{\mathrm{ext}} \text{ or } m\mathbf{a}_c = \mathbf{F}^{\mathrm{ext}}$$

where $\mathbf{a}_c = \mathrm{d}\mathbf{v}_c/\mathrm{d}t$ is the acceleration of the centre of mass. Thus, the centre of mass (centre of inertia) of a mechanical system travels like a particle whose mass is equal to the mass of the whole system and which is subject to a force equal to the principal vector of the external forces applied to the system. If the system being considered is a rigid body that travels with translational motion (1.5.1), the velocities $\mathbf{v}_i$of all the points of the body and $\mathbf{v}_c$ of its centre of mass are the same and are equal to velocity $\mathbf{v}$ of the body. Correspondingly, the acceleration of the body $\mathbf{a} = \mathbf{a}_c$ and the *basic equation in the dynamics of translational motion of a rigid body* is of the form $m\mathbf{a} = \mathbf{F}^{\mathrm{ext}}$.

2.6 Motion of a Body of Variable Mass

2.6.1

In Newtonian mechanics the mass of a body can be changed only if particles of matter are detached from or added to it. An example of such a body is a rocket. In its flight the mass of the rocket is gradually reduced because the gaseous products of combustion in the rocket engine are ejected through the nozzles.

The equation of translational motion of a body of variable

mass (Meshchersky equation) is

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{F}^{\mathrm{ext}} + (\mathbf{v}_1 - \mathbf{v})\frac{\mathrm{d}m}{\mathrm{d}t}$$

where m and $\mathbf{v}$ are the mass and velocity of the body at the instant of time being considered, $\mathbf{F}^{\mathrm{ext}}$ is the principal vector of the external forces (2.5.2) acting on the body, and $\mathbf{v}_1$ is the velocity of the detracted particles *after separation* from the rocket (if $\mathrm{d}m/\mathrm{d}t < 0$) or the added particles *before being joined* (if $\mathrm{d}m/\mathrm{d}t > 0$).

2.6.2

The second member on the right-hand side of the Meshchersky equation represents the additional force acting on a body of variable mass. It is called the *reactive force*:

$$\mathbf{F}_{\mathrm{r}} = (\mathbf{v}_1 - \mathbf{v})\frac{\mathrm{d}m}{\mathrm{d}t} = \mathbf{u}\frac{\mathrm{d}m}{\mathrm{d}t}$$

where $\mathbf{u} = \mathbf{v}_1 - \mathbf{v}$ is the relative velocity of the particles being separated from or joined to the body, i.e. their velocity with respect to a reference frame in translational motion together with the body.

The reactive force characterizes the mechanical action exerted on the body by the particles separated from or joined to it (for instance, the effect of the ejected exhaust gases on the rocket).

2.6.3

The equation of motion of a rocket not subject to the action of external forces is

$$m\frac{\mathrm{d}\mathbf{v}}{\mathrm{d}t} = \mathbf{u}\frac{\mathrm{d}m}{\mathrm{d}t}$$

If the initial velocity of the rocket is zero the rocket flies in a straight line in the direction opposite to that of the relative velocity $\mathbf{u}$ of the exhaust gases ejected from the engine nozzle. Here

$$m\frac{\mathrm{d}v}{\mathrm{d}t} = -u\frac{\mathrm{d}m}{\mathrm{d}t}$$

and at $\mathbf{u}$ = const the relation between the velocity of the rocket and its mass is expressed by *Tsiolkovsky's formula*

$$v = u\ln\frac{m_0}{m}$$

where m_0 is the initial, or launching, mass of the rocket.

2.6.4

The maximum velocity that a rocket can reach in the absence of external forces is called its *characteristic velocity*. This velocity is acquired at the moment the rocket engine stops because it has used up all the fuel and oxidizer that were on board,

$$v_{\max} = u\ln\frac{m_0}{m_0 - m_\mathrm{f}}$$

where m_f is the initial mass of the fuel and oxidizer. The effects of the earth's gravity and air resistance lead to an

appreciable reduction in the maximum velocity actually acquired by the rocket, during the operation of its engine, as compared to the characteristic velocity.

2.6.5

The *characteristic velocity of a multistage* (combination) *rocket* is

$$v_{\max} = \sum_{i=1}^{n} u_i \ln \frac{m_{0i}}{m_{0i} - m_{fi}}$$

where n is the total number of stages of the rocket, m_{fi} is the mass of the fuel and oxidizer intended for the operation of the ith stage engine, u_i is the exhaust gas velocity (nozzle velocity) of the ith stage engine, and m_{0i} is the launching mass of the multistage rocket, including all the stages from the ith to the nth. The velocity increase of a multistage rocket, compared to that of a single-stage one having the same launching mass and the same fuel and oxidizer capacity, is due to the reduction in the mass of the rocket as the first, second and subsequent stages are consecutively jettisoned when their fuel has been exhausted.